

1s		1s
2s		2p
3s		3p
4s	3d	4p
5s	4d	5p
6s	5d	6p
7s	6d	7p
	4f	
	4d	

Unit 4 -The Electron

Essential Knowledge and Guided Notes

Read Textbook: CK Chemistry Chapter 5: Electrons in Atoms
<https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/>

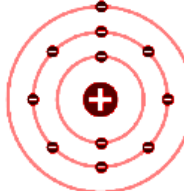
Watch Helpful Videos and Practice: Posted on CANVAS

Quick Electron Configuration Practice: <http://tinyurl.com/geekconfig>

Highlight Essential Knowledge:

A HISTORY OF THE ATOM: THEORIES AND MODELS

How have our ideas about atoms changed over the years? This graphic looks at atomic models and how they developed.

SOLID SPHERE MODEL	PLUM PUDDING MODEL	NUCLEAR MODEL	PLANETARY MODEL	QUANTUM MODEL
				
JOHN DALTON	J.J. THOMSON	ERNEST RUTHERFORD	NIELS BOHR	ERWIN SCHRÖDINGER
 1803	 1904	 1911	 1913	 1926

Big Idea #1: Our idea of the atom evolved: (See Reading #1 in the U4 Problem Set)

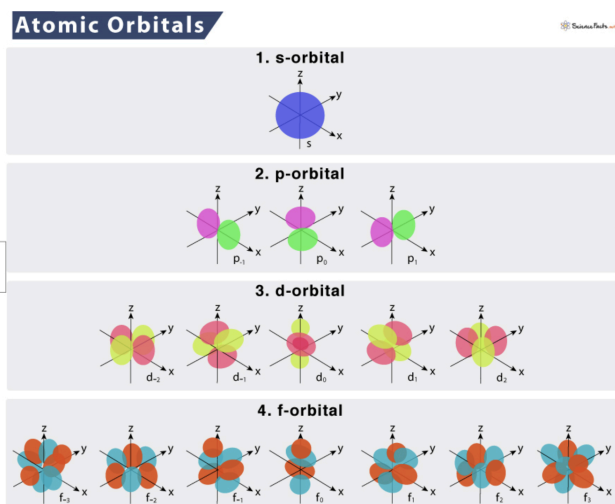
1. **Dalton's Model (1803):** Atoms are indivisible, indestructible particles.
2. **Thomson's Plum Pudding Model (1897):** Atoms are positively charged spheres with negatively charged electrons scattered throughout. He used **a cathode ray tube** to show this.
3. **Rutherford's Nuclear Model (1911):** Atoms have a small, dense nucleus containing positively charged protons, with electrons orbiting around the nucleus. His famous experiment is called the **gold foil experiment**.
4. **Bohr's Planetary Model (1913-1923):**
 - a. Special Orbits and Energy Levels: Bohr proposed that electrons orbit the nucleus in up to 7 specific energy levels.
 - b. Quantized Energy Levels: Electrons can move to higher energy orbits if given the right amount of energy.
 - c. Limitations: Unable to explain the behavior of electrons in elements larger than Hydrogen.

5. **Quantum Mechanical Model (1926):** Atoms consist of a nucleus with protons and neutrons, surrounded by electron clouds representing areas of electron probability. Ongoing research refines our understanding of atomic structure and behavior. Quantum mechanics underpins modern chemistry, physics, and technology, shaping our understanding of matter at the atomic level.
 - a. **Wave-Particle Duality:** Louis de Broglie's idea that electrons exhibit both wave and particle properties.
 - b. **Heisenberg's Uncertainty Principle** It's impossible to know both the exact position and momentum of a particle simultaneously.
 - c. **Schrodinger's Contributions:** Developed mathematical equations describing electron behavior, leading to the concept of orbitals. **Orbitals:** Electrons are found in high-probability regions (orbitals) around the nucleus, not in fixed paths. Electrons move in areas of probability, not fixed orbits like planets around the sun. Schrodinger's equations became the basis of quantum mechanics, explaining electron behavior and winning him a Nobel Prize.

Big Idea #2: Electron configurations and orbital diagrams show the arrangement of electrons around the nucleus of an atom based on their energy. (See Reading #2 in the U4 Problem Set)

1. **Principal Energy levels** are designated $n = 1-7$. The higher the number, the higher the energy and the farther away from the nucleus.
2. **Sublevels** within these principal energy levels are designated s, p, d, and f according to their shapes and relate to the regions of the Periodic Table. Each sublevel can hold a varying number of orbitals as seen in the table below.

Sublevel	Shape	Maximum # of Electrons per sublevel	# of Orbitals per sublevel
s	Sphere	2	1
p	Dumb-bell	6	3
d	Clover	10	5
f	Flower	14	7

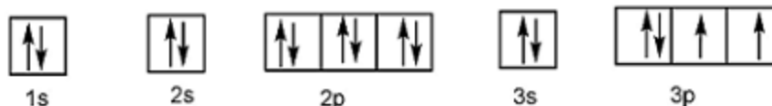


- Electrons are added one at a time to the lowest energy levels first (**Aufbau Principle**).
- Electrons occupy equal-energy orbitals, so a maximum number of unpaired electrons results (**Hund's Rule**).
- An orbital can hold a maximum of two electrons of opposite spins (**Pauli Exclusion Principle**).
- Electron Configurations** show what levels and sublevels are occupied.
- Orbital diagrams** show the orbitals themselves.

Here is the electron configuration and orbital diagram of a neutral Sulfur atom (atomic number 16):

Electron Configuration: **$1s^2 2s^2 2p^6 3s^2 3p^4$ or $[\text{Ne}] 3s^2 3p^4$**

Orbital Diagram:



- Atoms can gain or lose electrons within the outer energy level. Electrons in outer or highest energy levels are known as **valence electrons**. The configuration above shows that Sulfur has **6 valence electrons** as defined by electrons in the highest unoccupied energy level ($n=3$).
- Atoms emit light when energy is absorbed and released. The frequencies of light an atom emits are related to changes in electron energies (**atomic emission spectrum**).

Guided Notes

U4 Lesson #1 Models of the Atom Through Time

Models of Atoms through Time: Dr. Brian Cox Video Questions

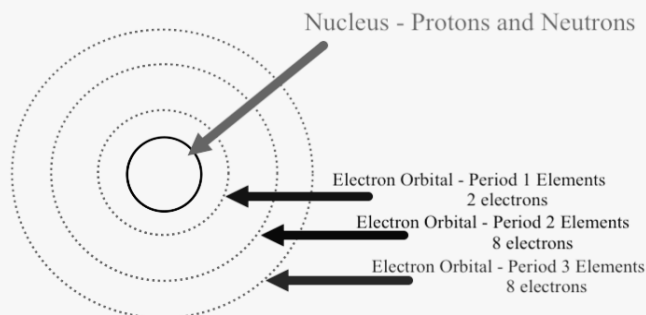
- 1860 (*Periodic Table*) - *Mendeleev* proposed the first _____ table with about _____ known elements.
- 1897 (Discovery that the atoms have parts: Electron) - *JJ Thompson* used a *cathode ray tube* to discover the first subatomic particle: _____ which was negatively charged. His model said the electrons were like the chocolate chips in a _____. *It was formerly known as the "plum pudding" model.*
- 1911 (Proposal of the Nucleus) - *Ernest Rutherford* used the decay of radioactive elements to produce beams of particles. Rutherford's desk is still _____ after 100 years as a result! Along with his colleagues, Geiger and Marsden, he fired radioactive alpha particles at thin _____ foil. They expected the particles would all pass right through. 1: _____ bounced back! This was known as the *Gold foil experiment*. The new nuclear model was like a _____ with the mass in the middle and the electrons orbiting like planets. In other words, the atom was mostly _____ space.
- 1932: (Protons and Neutrons) *Rutherford and James Chadwick* continued to probe the nucleus and discovered positive _____ and neutral _____. *It was formerly known as Rutherford's Nuclear Model.*
- Did protons, neutrons, and electrons explain everything? _____ What particle was later discovered to make up protons and neutrons? _____

U4 Lesson #2 - Quantum Mechanical Model and Atomic Orbitals

1. (1913) Bohr Model

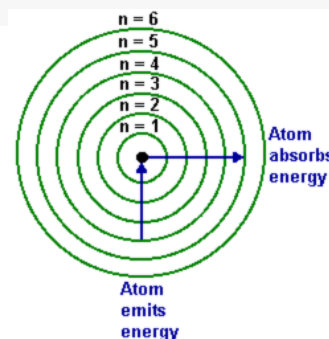
- It was known as the _____.
- Electrons are found in an _____ around the _____.
- Electrons have _____ that correspond to their _____.

Bohr's Atom



2. Quantum Levels in the Bohr Model

- An _____ can move from one level to the next by _____ or _____ just the right amount of _____.
- _____ : the amount of energy required to move an electron from one level to the next.
- The amount of energy an electron needs to gain or lose to move a level is _____.
- Some levels are more similar in energy (close together) and some are far apart.
- As an electron moves away from the nucleus the energy levels get _____ together.
- Was Bohr's model correct? _____
- It didn't work mathematically for elements larger than _____ (atoms with more than 1 electron!)



3. Quantum Mechanical Model

- It was derived by Erwin Schrodinger (1887-1961).
- As with all science it was based on the work of other scientists:
 - Louis de Broglie and his idea that electrons possess both wave and particle properties known as _____
 - Werner Heisenberg and his _____ that said it was impossible to know both the exact position and momentum of a particle simultaneously.
- It was a set of _____ that describes how electrons move in an atom.
- It was built on the Bohr model but was not the same.
- Electrons move in _____ (areas of probability like clouds) versus orbits (exact paths)

Bohr Model	Q.M.M.
Electrons exist in discrete energy levels	
Electrons move in a specific path around the nucleus	The path of the electron is determined by statistical probability
Electron Path (Orbits)	Electron Cloud/Orbitals

4. The Electron Cloud

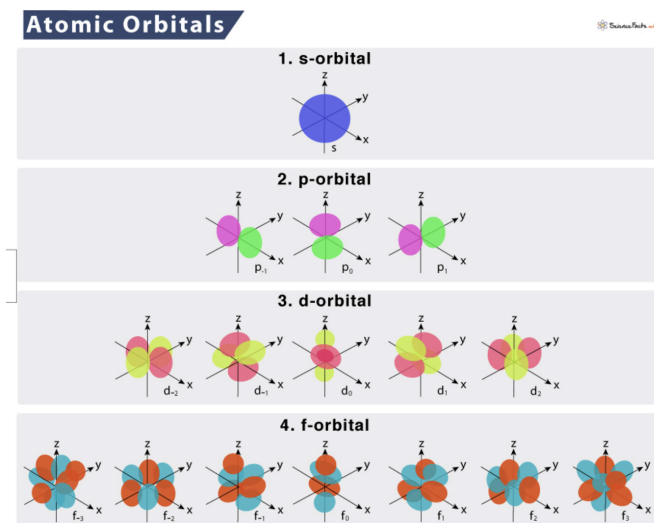
- Represented as a _____ where an electron _____ be.
- The cloud is more dense where the probability is _____, and less dense where the probability is _____.
- Each atom has several clouds or _____ corresponding to electrons possessing different energies (different energy levels)

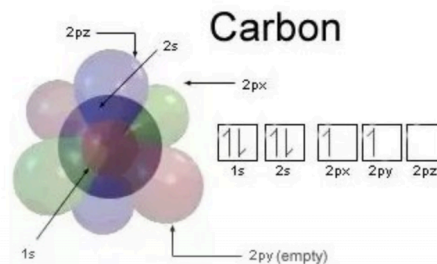
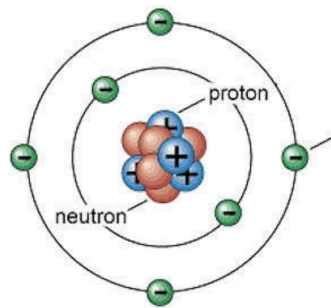
5. Atomic Orbitals

- The energy levels of electrons are labeled by a principal _____ number (n) = 1, 2, 3, 4...
- Each principal energy level has _____ designated by letters: s, p, d, f.
- Each sublevel has 1 or more _____ or regions of space where there is a high probability of finding an electron.
- Each orbital can hold 2 electrons of _____ spins

6. Summary Table:

Sublevel	Shape	# of orbitals per energy level	Max # of Electrons Per energy level
s	Sphere		
p	Dumbbell		
d	Clover		
f	Flower		





U4 Lesson #3 - Atomic Spectra Demonstration

Safety: Do not touch the spectrum tube or any part of the transformer or look directly at the light.

Pre-Demonstration Notes:

1. The quantum mechanical model grew out of the _____.
2. The amount of energy that light possesses depends on its _____ (v) and its _____ (λ)
3. The wavelength and frequency of light are _____ proportional to each other and when multiplied equal to the speed of light: $c = \lambda \nu$
 $c = \text{_____ m/s}$
4. The energy of light is directly proportional to frequency and inversely proportional to wavelength using this equation. $E = h\nu$ Planck's Constant (h) = $6.626 \times 10^{-34} \text{ J s}$
5. _____: Different forms of light have different energy based on their wavelength and frequency.
6. In 1913 Niels Bohr came up with an idea to explain why negative electrons didn't fall into the positive nucleus proposed by Rutherford by putting them in _____ like planets around the Sun.
7. He had this idea when he looked at the spectrum produced by gaseous atoms through a spectroscope.
8. In order to stay in "orbits" the electrons had to possess _____.
9. Spectroscopes break up light into its _____. Different wavelengths have different _____.
10. White light has a continuous _____. Red light has longer wavelengths (<650 nm) and thus less energy than violet light (>500 nm).
11. Analysis of light from a tube of glowing hydrogen emitted a _____ spectrum. It was not a continuous spectrum.
12. When electrons absorb a quantum of energy, they become "excited" and move to a higher energy level or _____. When they release energy, they move back down to their lowest energy level or _____. The released energy was in the form of a specific wavelength of _____ light.

13. Electrons move to lower energy levels by emitting light.
14. Light emitted from atoms is made up of only certain specific wavelengths of light.
15. Each atom has its own spectrum.

16. Atomic Spectra are like “fingerprints”.

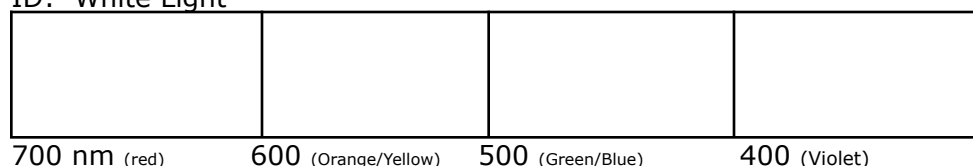
Summary: White light includes all energies of light and all wavelengths. It is continuous. Light from elements only contains certain wavelengths of light. The ground state is when electrons are in the lowest energy levels. The excited state is when electrons have absorbed energy and have jumped to higher energy levels temporarily. Electrons in the ground state become excited and then when they fall back to the ground state they release energy in the form of visible light. Emission spectra show that elements have discrete energy levels because they emit only certain wavelengths of light. The color of the light corresponds to the unique energy levels of that particular element. The pattern from each element is like a fingerprint that is useful for identification.

Demonstration Notes: Begin by observing a ‘white’ light source through the spectroscope. Look out the window of the classroom. Look through the small end of the spectroscope and point the slit at the wide end of the triangle toward the light source. You should see a rainbow of colors reflected on a rectangle matching the one in the diagram in your data section. Color in the data rectangle to record what you see.

The wavelengths of visible light are:

- **Violet:** 380–450 nm (688–789 THz frequency)
- **Blue:** 450–495 nm
- **Green:** 495–570 nm
- **Yellow:** 570–590 nm
- **Orange:** 590–620 nm
- **Red:** 620–700 nm (400–484 THz frequency)

ID: White Light

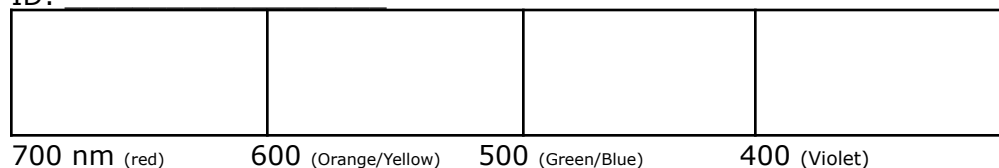


In turn with your group, you will observe the light produced by a spectrum tube of a single element. Line up the light with the slit in the spectroscope. While waiting your turn, sketch the spectrum of two of the elements from the slide show: Try to put the colors of visible light in their approximate wavelengths:

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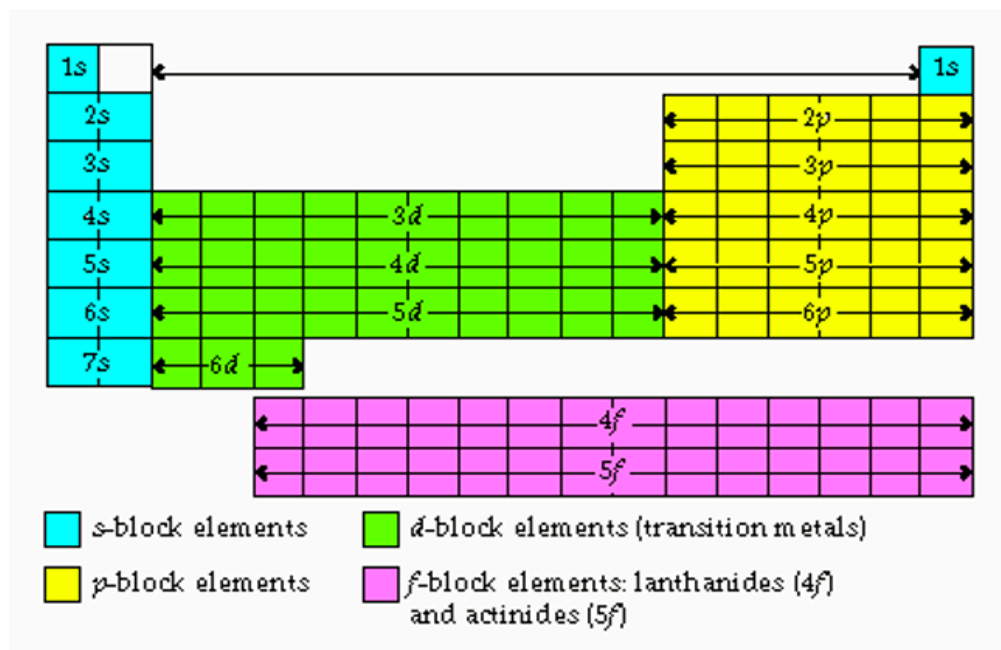


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U4 Lesson #4 Electron Configuration

1. Electrons interact with the nucleus to make the _____ arrangement possible. This usually is the lowest energy possible. It is called the _____ state.
2. Electrons are placed in orbitals in a specific pattern that is _____ than the simple Bohr orbits.
3. _____ rules dictate how electrons are configured: Aufbau Principle, Pauli Exclusion Principle, Hund's Rule
4. Think of the Periodic Table as a hotel with rooms. There are 7 floors (large numbers: 1,2,3,...) called "energy levels" and 4 types and numbers of rooms in parts of the hotel of various shapes called "sublevels" (letters: s,p,d,f). Now you must learn the hotel layout to place the electrons in their proper places. Use your "cheat sheet" for now:



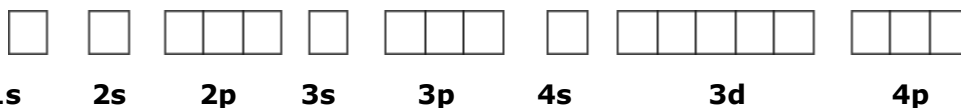
5. Three General Rules to Memorize when assigning electrons to orbitals

Name	Rule
	Electrons fill the lowest energy orbitals first. "Lazy Tenant" Rule. As Above: 1s,2s,2p,3s,3p,4s,3d,4p,_____
	Correct 1s Incorrect 1s Each orbital can hold TWO electrons with opposite spins represented by up and down arrows.
	✓ ✗ Within a sublevel place one electron per orbital before pairing them. "Empty Bus Seat" Rule

6. There are two ways to show where all the electrons are in your atom.

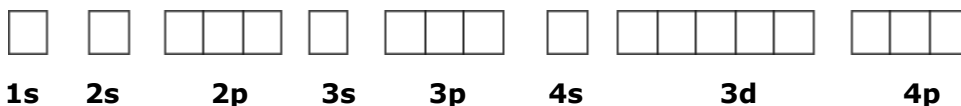
- **Electron Orbital Diagrams** show the actual orbitals.
- **Electron Configurations** are shorter and show just how many electrons are on each level and sublevel.

Oxygen (8 electrons): Orbital Diagram



Electron Configuration: _____

Sulfur (16 electrons): Orbital Diagram



Electron Configuration: _____

Noble Gas (short-hand) Notation for Sulfur: _____

Define Valence Electrons:

Label the Core Electrons and Valence Electrons for Sulfur: $1s^2 2s^2 2p^6 3s^2 3p^4$

Germanium (32 electrons):

Full Electron Configuration: _____

Noble Gas Notation: _____

Valence Electrons: _____

Let's do a few more examples:

Element (Atomic #)	Electron Configuration (May use noble gas shortcut)	Highest principal energy level	Number of Valence Electrons

